

Electronic supplementary material

Payoff scale reshapes how language models play
the prisoner’s dilemma

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S1 Scope of the corpus

The analysis in the main text uses one corpus of language-model transcripts, `Dataset/data_fairgame_frontier_llm`, and one synthetic corpus, `Dataset/noise_dataset`, which trains the LSTM branch of the strategy read-out and contains no language-model output at all. No other dataset in the repository enters any number reported in the paper.

The transcript corpus is a complete crossing of ten payoff scales, five models, five prompt languages, four persona pairings and ten replicates, at ten rounds per game and two agents per dyad. Table S1 gives its composition. Every one of the 250 model-by-language-by-scale cells holds exactly 40 dyads per language, or 200 per cell of the coarser model-by-scale grid, and the ingest raises an error rather than writing its tables if that is not what is on disk. It also checks that every game is ten rounds long, and that the payoff scale recovered from the recorded payoffs agrees with the scale the cell was run at, which catches a mislabelled directory.

S2 Cooperation, by scale, model and language

Table S2 gives the cooperation rate behind Fig. 1b of the main text, with bootstrap intervals. Table S3 breaks the same quantity down by prompt language and is the table behind Fig. 2.

The final column of Table S3 is the quantity we draw attention to in the main text: the range of cooperation across the ten payoff scales, computed within one model and one language.

It runs from 0.080 (Gemini 3.1 Flash-Lite in Arabic) to 0.449 (GPT-5.4 Nano in English). The sensitivity of a model to a transformation that changes nothing about the game is therefore not a single number even for a fixed model; it depends on the language the game was posed in.

S3 Variance decomposition

Table S4 is the full analysis-of-variance table summarised in the main text. It is computed on the 250 cell means rather than on the 20,000 agent-games, because the design is exactly balanced at the cell level, which makes the type-II decomposition order-independent.

The reading that matters for benchmark practice is the contrast between rows. The payoff scale carries 0.7% of the variance across cells as a main effect and is not significant ($p = 0.36$). Its interaction with the model carries 6.5% and its interaction with the prompt language 3.8%, and both are significant. A design that estimates a main effect for the payoff scale, pooling over models and languages, is therefore built to return exactly the null result it would return if no model responded to the scale at all.

S4 Persona

Table S5 gives the persona effect at every scale and model, which is the table behind Fig. 3a. The two leftmost columns are the scales at which every payoff printed is at most 1, since with base payoffs 0, 2, 6, 10 the largest number shown is 10λ .

Two features are worth stating explicitly because they are easy to misread from the figure. First, the sign reversal is a reversal of the *persona* effect, not of the cooperation rate: below the boundary the three reversing models cooperate more when told they are selfish than when told they are cooperative. Second, Qwen3 235B-A22B is inverted at every scale rather than only below the boundary, cooperating 0.072 under the cooperative persona against 0.798 under the selfish one. We report it because excluding it would flatter the analysis, but the paper draws no conclusion from the inversion itself beyond noting that its magnitude, like everything else measured here, depends on the payoff scale.

S5 Strategy read-out

S5.1 What the two branches do

Each agent-game is encoded as a sequence of two-letter tokens, one per round, pairing the previous round’s joint outcome with the action taken in the current round. The alphabet is shared between the synthetic training corpus and the language-model transcripts, which is what makes a classifier trained on the former applicable to the latter without any adaptation.

The rule base asks an exact question: is this trajectory precisely what AllC, AllD, Tit-for-Tat or Win-Stay-Lose-Shift would have played, given the history that actually occurred? Because all four rules are memory-one, the answer is a per-token lookup and requires no learning. It returns a set rather than a single label, because over ten rounds several canonical rules routinely prescribe the same moves. A player cooperating in all ten rounds against a cooperator is exactly consistent with three of the four rules at once, AllC, Tit-for-Tat and Win-Stay-Lose-Shift, and no method, learned or otherwise, can separate them on that evidence. Reporting the set is the correct answer, not a failure.

The LSTM ranks within whatever set the rule base returns, and decides alone only where the set is empty. It therefore cannot overturn a deduction; it can only break a tie the evidence genuinely leaves open, or supply an attribution where no canonical rule applies.

S5.2 Provenance, and which claims depend on the classifier

Over the whole corpus, 23.0% of agent-games are matched exactly by one canonical rule, 13.2% by several at once, and 63.7% by none. Those three provenances are reported separately throughout, and the claims in the main text are ordered by how much they depend on the learned component:

- The share of agent-games matched exactly by one rule (Table S6) and the number of rounds violating the nearest canonical rule are computed by exact matching. Neither depends on the LSTM in any way. These carry the main text’s strategy claim.
- The compositional claim, that the mix moves from unconditional towards reciprocal play (Table S7), uses the labels, and therefore leans on the LSTM for the 76.9% of agent-games that are ambiguous or unmatched. It is the weakest of the three and is presented as such.

Table S8 shows that this dependence is very unevenly spread across the four labels, which bounds how the compositional result may be read. AllC and AllD carry deduced shares of 26.0% and 32.7%, so a movement in those two is anchored in part by exact matching. Tit-for-Tat and Win-Stay-Lose-Shift are 95.4% and 97.3% unmatched, meaning the label is an attribution the LSTM makes about which canonical rule the play most resembles, on trajectories that follow none of them exactly. A rise in the Tit-for-Tat share is therefore evidence that play is drifting towards the reciprocal region of the space, and is not evidence that any model is implementing Tit-for-Tat. The main text states this where the result is reported.

S5.3 Validation

Table S9 reports the accuracy of the LSTM branch. Training uses the noise-free and 5% execution-noise variants of the synthetic corpus together, which forces the network to identify a rule from imperfect behaviour rather than from a clean signature; the 10% and 20% variants are held out entirely and measure how the read-out degrades as play departs further from any rule. Accuracy falls from 98.5% in distribution to 93.6% and 81.1% at those two unseen noise levels, which is the honest bound on how far the label distributions in Table S7 should be pushed.

S6 Why the model curves cancel under pooling

Table S10 gives the pairwise correlations between the five models’ cooperation-against-scale curves. Six of the ten pairs are negative, the mean is -0.10 , and the extremes are -0.72 and $+0.73$. Averaging curves that disagree in shape this strongly reduces the range from a mean of 0.136 within a model to 0.048 for the average, a factor of 2.8 . The cancellation is arithmetic rather than statistical, and no increase in sample size affects it.

S7 Reproduction

The pipeline runs in order and each stage writes its outputs under `Analysis/scaling/`:

- `s00_build.py` parses the transcripts into `rounds.parquet` and `games.parquet`, and refuses to write either if the grid is incomplete.
- `s01_train_lstm.py` trains the LSTM branch on the synthetic corpus and writes `models/strategy_lstm.pt`.
- `s02_readout.py` applies the hybrid read-out to all 20,000 agent-games and writes `readout.parquet`.
- `s03_stats.py` and `s04_strategy_stats.py` compute every statistic in the paper into `tables/`.
- `s05_figures.py` draws Figs. 1 to 5.
- `s06_tables.py` and `s07_supplementary.py` generate the LaTeX tables in the main text and in this supplement.

Random seeds are fixed: 1234 for classifier training, 20260905 for the bootstrap and the permutation test.

S8 Supplementary tables

Table S1: The corpus, by model. Every model was run on the same ten payoff scales and five prompt languages. “Cell n ” is the number of dyads in each model $\times$ language $\times$ scale cell; the design is balanced with none missing, and the build refuses to write its tables otherwise.

Model	Scales	Lang.	Dyads	Agent-games	Decisions	Cell n	Coop.
Claude Haiku 4.5	10	5	2,000	4,000	40,000	40	0.415
GPT-5.4 Nano	10	5	2,000	4,000	40,000	40	0.606
Gemini 3.1 Flash-Lite	10	5	2,000	4,000	40,000	40	0.617
Gemini 3.5 Flash-Lite	10	5	2,000	4,000	40,000	40	0.751
Qwen3 235B-A22B	10	5	2,000	4,000	40,000	40	0.435
All	10	5	10,000	20,000	200,000	40	0.565

Table S2: Cooperation by model and payoff scale. Each entry aggregates 200 dyads. The bracketed line under each model is a 95% confidence interval from a bootstrap that resamples whole dyads.

Model	$\lambda=0.01$	$\lambda=0.1$	$\lambda=0.25$	$\lambda=0.5$	$\lambda=1$	$\lambda=2$	$\lambda=5$	$\lambda=10$	$\lambda=100$	$\lambda=1000$
Claude Haiku 4.5	0.407 [0.37,0.45]	0.379 [0.34,0.42]	0.371 [0.34,0.40]	0.427 [0.39,0.46]	0.452 [0.42,0.49]	0.444 [0.41,0.48]	0.424 [0.39,0.46]	0.435 [0.40,0.47]	0.407 [0.38,0.43]	0.401 [0.37,0.43]
GPT-5.4 Nano	0.438 [0.40,0.48]	0.540 [0.50,0.58]	0.644 [0.61,0.68]	0.598 [0.56,0.63]	0.576 [0.54,0.61]	0.637 [0.60,0.67]	0.680 [0.65,0.71]	0.633 [0.60,0.67]	0.663 [0.64,0.69]	0.653 [0.62,0.68]
Gemini 3.1 Flash-Lite	0.641 [0.61,0.67]	0.648 [0.62,0.68]	0.621 [0.59,0.65]	0.640 [0.61,0.67]	0.635 [0.61,0.67]	0.614 [0.58,0.65]	0.614 [0.58,0.65]	0.598 [0.56,0.63]	0.584 [0.54,0.62]	0.577 [0.54,0.61]
Gemini 3.5 Flash-Lite	0.792 [0.74,0.84]	0.694 [0.64,0.74]	0.872 [0.85,0.89]	0.800 [0.77,0.83]	0.774 [0.75,0.80]	0.710 [0.67,0.74]	0.697 [0.66,0.73]	0.754 [0.72,0.78]	0.734 [0.70,0.77]	0.682 [0.64,0.72]
Qwen3 235B-A22B	0.476 [0.43,0.52]	0.494 [0.44,0.54]	0.455 [0.41,0.50]	0.432 [0.39,0.47]	0.416 [0.38,0.45]	0.427 [0.39,0.46]	0.419 [0.38,0.45]	0.422 [0.39,0.45]	0.399 [0.37,0.43]	0.411 [0.38,0.44]

Table S3: Cooperation by model, prompt language and payoff scale. 40 dyads per entry. The final column is the range across the ten scales for that model and language; it runs from 0.080 to 0.449, so the scale sensitivity of a model is a property of the model and the language jointly.

Model	Language	0.01	0.1	0.25	0.5	1	2	5	10	100	1000	Range
Claude Haiku 4.5	English	0.35	0.40	0.32	0.42	0.44	0.45	0.41	0.42	0.45	0.46	0.136
	French	0.65	0.47	0.50	0.33	0.37	0.43	0.41	0.40	0.38	0.40	0.314
	Vietnamese	0.17	0.17	0.28	0.29	0.26	0.20	0.30	0.22	0.23	0.23	0.130
	Chinese	0.53	0.52	0.45	0.61	0.69	0.57	0.56	0.66	0.58	0.56	0.245
	Arabic	0.35	0.33	0.31	0.48	0.51	0.56	0.43	0.47	0.40	0.36	0.249
GPT-5.4 Nano	English	0.45	0.64	0.85	0.88	0.86	0.90	0.88	0.87	0.89	0.76	0.449
	French	0.75	0.78	0.78	0.57	0.55	0.60	0.62	0.67	0.55	0.74	0.226
	Vietnamese	0.49	0.46	0.58	0.47	0.50	0.62	0.75	0.54	0.65	0.63	0.289
	Chinese	0.24	0.45	0.65	0.60	0.49	0.57	0.61	0.58	0.62	0.65	0.414
	Arabic	0.25	0.37	0.37	0.48	0.47	0.49	0.54	0.51	0.61	0.49	0.360
Gemini 3.1 Flash-Lite	English	0.74	0.62	0.67	0.62	0.63	0.62	0.59	0.62	0.57	0.62	0.167
	French	0.57	0.66	0.70	0.63	0.65	0.60	0.60	0.61	0.60	0.56	0.145
	Vietnamese	0.58	0.56	0.54	0.64	0.57	0.55	0.57	0.49	0.47	0.47	0.170
	Chinese	0.62	0.66	0.53	0.61	0.59	0.60	0.60	0.58	0.58	0.58	0.133
	Arabic	0.69	0.74	0.66	0.70	0.74	0.69	0.71	0.69	0.69	0.66	0.080
Gemini 3.5 Flash-Lite	English	0.94	0.66	0.92	0.84	0.74	0.59	0.53	0.65	0.64	0.60	0.412
	French	0.91	0.63	0.86	0.72	0.78	0.63	0.75	0.78	0.59	0.61	0.317
	Vietnamese	0.72	0.63	0.93	0.86	0.81	0.76	0.62	0.64	0.89	0.73	0.314
	Chinese	0.69	0.80	0.76	0.64	0.70	0.73	0.67	0.76	0.66	0.73	0.162
	Arabic	0.70	0.75	0.89	0.94	0.83	0.84	0.92	0.93	0.89	0.74	0.242
Qwen3 235B-A22B	English	0.47	0.51	0.53	0.44	0.42	0.49	0.43	0.50	0.37	0.46	0.155
	French	0.41	0.40	0.35	0.39	0.37	0.36	0.35	0.35	0.34	0.33	0.083
	Vietnamese	0.52	0.55	0.48	0.48	0.45	0.45	0.47	0.44	0.40	0.42	0.150
	Chinese	0.48	0.52	0.46	0.44	0.43	0.45	0.45	0.44	0.49	0.47	0.084
	Arabic	0.50	0.50	0.46	0.42	0.40	0.40	0.40	0.38	0.39	0.38	0.124

Table S4: Variance decomposition of the cell means. Type-II analysis of variance on the 250 model $\times$ language $\times$ scale cell means, where the design is exactly balanced. The payoff scale has no significant main effect while both of its interactions are significant.

Term	Sum sq.	df	<i>F</i>	<i>p</i>	% variance
Model	3.9188	4	206.80	< 0.001	56.5
Language	0.2092	4	11.04	< 0.001	3.0
Payoff scale	0.0471	9	1.10	0.364	0.7
Model $\times$ language	1.3665	16	18.03	< 0.001	19.7
Model $\times$ scale	0.4542	36	2.66	< 0.001	6.5
Language $\times$ scale	0.2615	36	1.53	0.041	3.8
Residual	0.6822	144			9.8

Table S5: Persona effect by model and payoff scale. The cooperation rate of an agent instructed that it is cooperative minus that of an agent instructed that it is selfish, at each scale. The two leftmost columns are the scales at which every payoff printed is at most 1.

Model	$\lambda=0.01$	$\lambda=0.1$	$\lambda=0.25$	$\lambda=0.5$	$\lambda=1$	$\lambda=2$	$\lambda=5$	$\lambda=10$	$\lambda=100$	$\lambda=1000$
Claude Haiku 4.5	-0.211	-0.232	+0.206	+0.200	+0.179	+0.261	+0.277	+0.158	+0.168	+0.211
GPT-5.4 Nano	-0.172	-0.098	+0.015	-0.026	-0.053	+0.032	+0.041	+0.055	-0.017	+0.100
Gemini 3.1 Flash-Lite	+0.552	+0.602	+0.640	+0.670	+0.693	+0.681	+0.707	+0.696	+0.659	+0.692
Gemini 3.5 Flash-Lite	-0.335	-0.578	+0.122	+0.234	+0.241	+0.258	+0.313	+0.203	+0.193	+0.230
Qwen3 235B-A22B	-0.901	-0.922	-0.811	-0.768	-0.702	-0.635	-0.690	-0.602	-0.657	-0.566

Table S6: Share of agent-games matched exactly by one canonical rule (%). Computed by exact rule matching, with no learned component. 400 agent-games per entry.

Model	$\lambda=0.01$	$\lambda=0.1$	$\lambda=0.25$	$\lambda=0.5$	$\lambda=1$	$\lambda=2$	$\lambda=5$	$\lambda=10$	$\lambda=100$	$\lambda=1000$
Claude Haiku 4.5	20.0	15.8	3.2	3.5	4.0	2.8	4.0	2.5	4.5	3.5
GPT-5.4 Nano	12.5	7.0	9.2	6.2	5.5	6.5	8.2	7.2	8.5	9.0
Gemini 3.1 Flash-Lite	16.5	19.0	21.2	23.5	29.0	23.0	30.0	29.5	27.5	26.8
Gemini 3.5 Flash-Lite	30.8	33.8	20.2	13.0	20.5	14.2	14.8	11.5	14.0	16.2
Qwen3 235B-A22B	77.0	67.2	63.0	62.7	59.8	45.2	58.0	43.2	55.0	42.0

Table S7: Strategy composition by model and payoff scale (%). The label assigned by the hybrid read-out: the rule base fixes the candidate set and the LSTM ranks within it, deciding alone only where no canonical rule fits. Columns sum to 100 within each model and scale.

Model	Label	0.01	0.1	0.25	0.5	1	2	5	10	100	1000
Claude Haiku 4.5	AllC	27.0	20.0	14.2	20.0	21.8	18.2	15.2	18.8	12.8	14.0
	TFT	20.5	23.5	23.5	18.0	16.8	16.5	19.5	17.8	20.8	22.2
	WSLS	8.8	13.5	15.2	13.5	20.2	21.2	20.5	18.2	21.8	19.0
	AllD	43.8	43.0	47.0	48.5	41.2	44.0	44.8	45.2	44.8	44.8
GPT-5.4 Nano	AllC	29.8	41.8	52.2	46.2	48.5	54.2	58.5	52.2	53.2	50.7
	TFT	11.8	11.5	10.5	11.8	10.2	13.2	13.8	13.2	13.0	17.8
	WSLS	12.5	12.5	14.5	15.5	13.8	10.5	12.5	10.2	15.5	14.0
	AllD	46.0	34.2	22.8	26.5	27.5	22.0	15.2	24.2	18.2	17.5
Gemini 3.1 Flash-Lite	AllC	55.8	57.0	58.2	61.0	59.8	55.0	58.5	53.8	52.2	52.2
	TFT	6.5	3.8	2.2	0.2	2.5	3.5	1.2	2.2	6.2	5.8
	WSLS	12.8	13.2	8.8	2.8	8.0	7.2	6.5	7.2	7.2	6.0
	AllD	25.0	26.0	30.8	36.0	29.8	34.2	33.8	36.8	34.2	36.0
Gemini 3.5 Flash-Lite	AllC	73.8	59.8	74.0	64.0	58.0	60.0	51.7	59.8	56.2	51.5
	TFT	6.8	2.2	10.2	16.2	17.0	16.2	15.2	13.5	15.2	14.2
	WSLS	6.2	13.2	14.0	16.2	21.2	12.0	18.0	18.5	16.5	15.2
	AllD	13.2	24.8	1.8	3.5	3.8	11.8	15.0	8.2	12.0	19.0
Qwen3 235B-A22B	AllC	42.0	47.8	39.2	38.0	36.8	36.5	35.5	35.2	34.5	33.8
	TFT	0.5	0.0	0.0	0.0	1.2	0.8	2.0	6.2	0.0	4.8
	WSLS	9.0	2.5	8.8	13.5	7.2	12.0	8.5	8.2	9.2	10.0
	AllD	48.5	49.8	52.0	48.5	54.8	50.7	54.0	50.2	56.2	51.5

Table S8: Provenance of each label. For every label, the share of the agent-games carrying it that were deduced (exactly one canonical rule fits), ambiguous (several fit, and the LSTM ranked within that set) or unmatched (no canonical rule fits, so the label is an attribution the LSTM makes alone). The asymmetry is severe and bounds how the compositional result may be read: AllC and AllD carry a substantial deduced share, whereas the TFT and WSLS labels are almost entirely attributions.

Label	n	Deduced (%)	Ambiguous (%)	Unmatched (%)
AllC	8,884	26.0	29.6	44.4
TFT	2,010	4.6	0.0	95.4
WSLS	2,494	2.0	0.7	97.3
AllD	6,612	32.7	0.0	67.3

Table S9: Validation of the LSTM branch. Trained on synthetic AllC / AllD / Tit-for-Tat / Win-Stay-Lose-Shift trajectories at zero and 5% execution noise, balanced at 40,320 sequences per strategy per level. The 10% and 20% noise levels are never trained on. No language-model transcript enters training.

Split	n	Accuracy
held-out (NoNoise + Noise005)	64,512	0.9846
OOD (Noise01)	161,280	0.9364
OOD (Noise02)	161,280	0.8107
held-out recall: AllC	16,150	0.9853
held-out recall: AllD	16,203	0.9952
held-out recall: TFT	16,193	0.9744
held-out recall: WSLS	15,966	0.9834

Table S10: Pairwise correlation between the model curves. Correlation of the ten-point cooperation-against-scale curves of each pair of models. Six of the ten pairs are negative and the mean is -0.10 , which is why averaging the curves cancels most of the movement.

	Claude Haiku	GPT-5.4 Nano	Gemini 3.1	Gemini 3.5	Qwen3 235B-A22B
Claude Haiku 4.5		+0.11	-0.03	-0.18	-0.56
GPT-5.4 Nano	+0.11		-0.70	-0.23	-0.72
Gemini 3.1 Flash-Lite	-0.03	-0.70		+0.36	+0.73
Gemini 3.5 Flash-Lite	-0.18	-0.23	+0.36		+0.22
Qwen3 235B-A22B	-0.56	-0.72	+0.73	+0.22	